

Embodied Mark Making: Reframing motion capture and drawing as disruptive and collaborative pedagogy

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Abstract

This paper investigates the integration of experimental motion capture into a collaborative drawing workshop as a pedagogical strategy to foster embodied learning and multimodal thinking in creative education. Facilitated by UNSW's Drawing Research Group and involving visiting academics and animation students from Hong Kong Polytechnic University, the workshop explored drawing as a time-based, spatially immersive process, extending traditional mark making through full-body movement, motion capture, and collective experimentation. Activities ranged from large-scale floor drawings to sensory tabletop exercises, scaffolded through creative prompts that encouraged play, improvisation, and non-verbal communication. Emphasising process over product, the workshop reframed motion capture as a generative material, subject to distortion, abstraction, and surprise, rather than a precision tool. Drawing on Bishop's *Artificial Hells* [Bishop, 2012] in which disruptive strategies for participatory art are prioritised, as well as embodied cognition, posthuman creativity, and Universal Design for Learning (UDL), the paper argues for a reimagining of authorship, assessment, and participation in art and design education. Findings highlight how multimodal, collaborative tasks foster student agency, critical engagement with digital materiality, and experimental learning literacy. The workshop affirms the value of interdisciplinary, sensorimotor-rich environments that blur boundaries between drawing, performance, and media art, modelling an inclusive and future-facing pedagogy.

CCS Concepts

• **Applied computing** → Education; Collaborative learning; • **Computing methodologies** → Computer graphics; Animation; Motion capture.

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1 Introduction

This paper explores the integration of experimental motion capture into a collaborative drawing workshop as a pedagogical strategy. Designed and implemented by researchers from University of New South Wales's (UNSW) Drawing Research Group (DRG), the researchers facilitated a group of 26 animation students to reconceive both drawing and motion capture as a full-body, time-based, and spatially immersive process. This reframing activates embodied learning and multimodal thinking, encouraging students to engage with movement, gesture, and spatial composition as core elements of mark making and experimentation. It included both group and individual exploration and activities. While conventional animation and drawing pedagogy understandably emphasises the finished artefact or technical mastery [Najafi et al., 2024], such models prioritise outcomes over the embodied, processual, and relational dimensions of learning. This project addressed that gap by foregrounding process over product, treating drawing and motion capture as iterative, sensory, and collaborative acts that expand how students think through making. By positioning embodiment as a critical learning strategy, the workshop facilitators challenged conventional hierarchies of skill and opened new pathways for inclusion, agency, and multimodal cognition.

The facilitators guided the activities via visual and verbal prompts, including “Mirror a partner's movement,” “Draw tension into the space,” or “Make an action that is meaningless,” to help scaffold the learning experience without constraining creative outcomes. Participants developed intuitive understandings of composition, space, gesture, and rhythm through trial, error, and peer observation, core to exploratory learning in creative media education. Central to the workshop was an investigation into how different bodies and motion qualities could be visualised and interpreted through multiple modalities of drawing and motion capture. Equally important was how unexpected moments of disruption emerged as productive forces within the collaborative setting. As Bishop [2012] argues, ‘unease, discomfort or frustration—along with fear, contradiction, exhilaration and absurdity—can be crucial to any work's artistic impact.’ One participant observed, the

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workshop enabled a 'Letting go of control', leading to unpredictable results. Collaborative art practice can disrupt participants from their comfort zones, and participants can likewise generate disruption as a creative act in and of itself. In this context, moments of divergence and spontaneous interventions were not treated as obstacles but reframed as conditions for innovation.

We combined motion capture with traditional drawing materials, charcoal, pen, and pencil, across activities from small sketches to large-scale floor works. Overhead video projection offered participants alternative perspectives, encouraging reflection on their embodied interactions. These outputs supported critical discussion of gesture, temporality, and representation across analogue and digital forms. Experimentation revealed motion qualities as trails, point clouds, or abstractions, where misread gestures and noise were treated not as errors but as prompts for dialogue on participation and material possibility, aligning with creative practice methods that value iteration and surprise. The workshop modelled a multimodal pedagogy that bridged drawing, animation, performance, and social practice. Its inclusive, playful and non-verbal modes of participation proved especially effective for exploring accessibility, embodied cognition, and collaborative storytelling. Through embodied experimentation, participants developed reflective practice, digital fluency, and creative risk-taking, core competencies for contemporary media arts education.

2 Background to Collaborative Approach

The project benefitted from the experience of colleagues at UNSW, the UNSW Drawing Research Group (DRG) as well as staff and students at Hong Kong Polytechnic University (PolyU). This collaboration builds on Karen Kriss's interdisciplinary art practice, which spans motion capture, 3D animation, drawing, and ceramics, and David Eastwood's extensive experience in painting and drawing. Kriss, with over 10 years industry experience and 15 years teaching computer graphics and motion capture, reframes the technology not just as an industrial tool but also as a medium for embodied inquiry, multimodal research, and inclusive participation, particularly through community projects co-designed with people with disability. Eastwood, drawing on 25 years as an educator and artist, emphasises equitable curriculum design, practical demonstration, and experimental processes that support diverse student needs. His teaching fosters agency, creativity, and reflection, encouraging students to take risks and engage deeply with artistic practice. Hossein Najafi has developed curricula bridging motion capture, animation, game design, and VFX, combining technical depth with creative exploration. Yuet Kai Chan contributes a multidisciplinary background in film, animation, and immersive media, highlighting motion capture's capacity to capture expressive detail. Jae-Eun Oh, an animation scholar and program leader, researches storytelling, nostalgia, and project-based learning, while Peter Sharp's long career in drawing and painting education emphasises disruptive, inclusive, and exploratory approaches. Together, the team fostered a dynamic environment where students engaged across disciplines, technologies, and practices.

The DRG at UNSW Art & Design approaches drawing as a dynamic, interdisciplinary practice that embraces experimentation,

embodiment, and emergent media. Its workshops challenge disciplinary silos by engaging diverse materials, sensory modes, and collaborative formats to reframe drawing as event-based and performative. Within this ethos, motion capture was embedded not as a technical tool but as a responsive material for mark making, performance, and play. Students and staff were invited to treat movement, gesture, and space as compositional elements, emphasising process over product and welcoming distortion, abstraction, and surprise. Conceived by Karen Kriss and David Eastwood, the workshop extended DRG's collaborative experiments in drawing-as-research, generating experiential knowledge beyond language [Roberts and Riley, 2014]. Building on de Zegher's [2011] definition of drawing as a "kinesthetic practice of traction," motion capture was adopted to record and express drawing as a kinetic medium. The visit of PolyU animation students provided the opportunity to share and test these experimental methods in an international, collaborative context.

The workshop began with an introductory presentation that outlined motion capture techniques, positioned the activities within contemporary drawing practice, and shared images of relevant artworks as both compositional elements and provocations. Participants then completed a hand-drawing exercise, sketching the gestures of each speaker during the presentation. This warm-up encouraged them to consider the relationship between drawing and motion, blending figurative representation with abstraction. The group then moved to large-scale floor drawing in two phases: the first focused on immersive, embodied compositions that traced bodies in motion, while the second invited new participants to respond directly to verbal prompts. The final activity was a multi-sensory tabletop drawing and dining experience, which further extended the workshop's exploration of embodied mark making and collaborative experimentation. The motion capture component employed an optical tracking system (10 x Vicon T40), using reflective markers placed on participants' wrists. During the workshop data was recorded at 120 fps and visualised in real time as both point cloud data and various character rigs via Blade and MotionBuilder. This component was discussed for integration after the workshop as analogue drawing reflection or generative possibilities for everyone involved. This methodological framing situates motion capture not as a precision-recording tool that prioritises the outcomes, but instead as a generative medium for embodied inquiry.

3 Playful Embodied Learning

Play was integral to creating a rich, multi-sensory experience for both staff and students during the workshop. We grounded activities in imaginative themes and playful scenarios, inviting participants to move and draw in response to verbal prompts. This emphasis on physical play aligns with theories of embodied cognition, which posit that cognitive processes are deeply rooted in the body's interactions with the world [Wilson, 2002]. Importantly, the educational value of play has also gained renewed emphasis within inclusive pedagogies such as Universal Design for Learning (UDL). The Center for Applied Special Technology (CAST) UDL Guidelines [CAST, 2018] now explicitly identify play and joy as critical components for fostering engagement, particularly in the newly articulated checkpoint 7.2: "Enhance relevance, value, and

authenticity.” This reflects a broader shift toward recognising adult play not just as motivational, but as central to deep learning, experimentation, social bonding and creative risk-taking. [CAST, 2018, Forbes, 2021] According to Tsai [2012] the effects of play in adult learners is underestimated.

In other words, learning occurs through the body: cognition is deeply dependent on sensorimotor experiences and active engagement with the environment [Barsalou, 2008]. By enacting scenarios (for example, “mirror a partner’s movement” or “make an action that is about light”), students weren’t just mentally planning responses, they felt and improvised them in real time, blurring the line between thinking and doing. Research in education supports this approach: embodied learning involves the whole body-mind, as learners make meaning through the senses, movement, and interaction with their surroundings [Kerka, 2002, Yakhlef, 2010]. As Juntunen [2020] argues, embodied learning is not automatic; it must be intentionally activated through tasks that engage learners’ bodies in meaningful ways. In our sessions, non-verbal modes of interaction, bodily gestures, expressive movements, improvised role-play, became powerful tools of communication and creative thinking. Such physical enactment allowed students to tap into tacit forms of knowledge and imagination that purely intellectual exercises might miss. By privileging play and movement alongside drawing, the workshop invited “embodied” ways of knowing that can enhance both creative ideation and memory retention, as supported by cognitive research on enactment and recall through gesture [Goldin-Meadow, 2010].

Critically, the playful embodied approach also encouraged multimodal thinking. Rather than treating drawing as a purely visual task, we integrated auditory cues, spatial movement, and tactile feedback, engaging multiple senses and modes of expression at once. Cognitive neuroscience has shown that learning through multiple modalities yields deeper understanding: when students engage more than one sense (for example, sketching while listening, or moving to express a concept), their brains form richer and more integrated memory networks [Shams and Seitz, 2008]. In one recent review, Mathias and von Kriegstein [2023] describe this as “multimodal enrichment”, the presence of complementary information across sensory or motor modalities can markedly benefit learning outcomes by literally strengthening neural connections across different brain regions [2023]. For instance, a comprehensive study by Mathias et al. [2021] found that children learning new vocabulary recalled words significantly better by over 70% in some cases when they performed meaningful gestures or whole-body movements while learning, compared to when they learned the words through listening or visuals alone.

In our workshop, the multimodal nature of the tasks clearly illustrated the pedagogical principle that engaging multiple sensory modalities can deepen creative learning. For example, the prompt to “draw tension into the space” required participants to translate a felt kinaesthetic sensation into a visual mark, literally blending bodily movement with graphic expression. This approach resonates with a lineage of experimental art practices that deliberately fuse drawing, sound, and movement. Conceptual artists anticipated this multimodal fusion: Two Stage Transfer Drawing [Oppenheim, 1971] by Dennis Oppenheim famously had the artist and his son draw on each other’s bodies, turning a drawing exercise into an

embodied, performative exchange. Contemporary performance and media artists, for instance, have created works where physical gestures become visual traces or where sound is rendered visible through mark making. The recent Performance on Paper exhibition [Takahatake, 2025] showcased how artworks on paper can “store sound and movement, becoming lasting visual records of ephemeral sonic and dance experiences,” including drawings produced as the physical trace of musical and dance performances.

Educators and researchers in arts-based learning similarly emphasise the value of such embodied, multimodal approaches. Research suggests that sensory-rich, non-verbal engagement can increase enjoyment and open students up to new possibilities in their environment [Freeman et al., 2014; Mavilidi et al., 2015]. By shifting the focus from purely mental effort to a holistic mind-body experience, we observed participants enter states of deep engagement or “flow” a term for the immersive, creative focus that often arises during playful physical involvement [Csikszentmihalyi, 1990]. In these moments of creative flow, students seamlessly integrated physical, social, and artistic competencies: they interacted with peers nonverbally, to translate imaginative ideas into bodily expression, and to take creative risks without fear of “being wrong.” As Veon [2023] states “Fostering risk-taking as an outcome has personal and professional benefits for students.” This convergence of modalities in a playful context nurtured an atmosphere of open exploration, where trial-and-error was welcomed.

Activating embodied learning through play proved to be a powerful pedagogical strategy in the workshop. It leveraged the innate human capacity for learning through movement and multi-sensory engagement, as documented by extensive research in cognitive science and education [Wilson, 2002; Barsalou, 2008; Juntunen, 2020]. The playful, physically immersive tasks not only made the session more enjoyable but also facilitated multimodal thinking. Students learned by seeing, hearing, moving, and drawing, leading to more integrated understanding. Equally important, this approach encouraged a reflective mindset toward the digital tools at hand. This resonates with practice-led art education approaches that value iteration, failure, and material exploration as essential to creativity and learning [Bolt, 2007; Sullivan, 2010]. Through playful embodied learning, the students experienced firsthand how the body can think and how technology can participate in the creative process, lessons that are likely to remain with them far beyond the workshop, as they continue to navigate and shape the evolving landscape of contemporary media arts education (Figure 1).

4 Multimodal Innovation and Student Agency

This project demonstrated the transformative potential of multimodal practice as both pedagogy and creative framework. By situating the workshop at the intersection of drawing, motion capture, performance, and digital media, students engaged in transdisciplinary knowledge-making that moved beyond medium-specific traditions [Nicolescu, 2002; Candy and Edmonds, 2018]. Animation students collaborated with drawing researchers and performers, treating technology not as a fixed workflow but as a material for thought: “These materials think in us, as we think through them.” [Ingold, 2013].



Figure 1: Image showing student and Drawing Research Group participation in the workshop. Credit: Cassandra Hannagan

The format also fostered student agency: rather than passive users, students became co-authors whose gestures and decisions shaped data and outcomes. Research shows that creative decision-making enhances motivation and ownership [Stefanou et al., 2004]. Motion capture underscored distributed authorship, with works emerging from constellations of human and nonhuman actors, resonating with post-human creativity [Bennett, 2010; Manning, 2009]. Bennett's "vibrant matter" and Manning's [2009] theorisation of assemblages frame the motion capture system, drawing and prompts as co-agents, shifting authorship toward a more-than-individual processual field.

This perspective displaced the demand for perfect outputs, valuing responsiveness, care, and iteration. It reframed failure and disruption as catalysts for reflection, cultivating experimental learning literacy, the ability to navigate ambiguity, work iteratively, and respond to material processes [Sullivan, 2010; Bolt, 2007; Candy and Edmonds, 2018]. Such competencies, diverse collaboration, fluency with technology, and conceptual agility are essential for future media arts practitioners [Koh and Abbas, 2015] alongside a "a 'lifelong kindergarten' spirit" [Gauntlett and Thomsen, 2013]. The workshop thus modelled an ecology of openness where students learned through doing, reflecting, and co-creating, underscoring the value of interdisciplinary practice.

5 Creative Outcomes and Drawing Connections

The workshop's outcomes extended beyond the immediate experience. We captured a top-down video of the students' collective motion and drawing process, providing an additional perspective on their work (Figure 2). This bird's-eye documentation became a reflective tool, allowing students to see the patterns of movement and mark making they had created together. Contextualising the motion-capture output in relation to drawing was crucial: we wanted participants to recognise how their bodily actions translated into visual forms, whether through digital trails or analogue media like charcoal and graphite. This was also later used as content analysis.

In this regard, the project draws inspiration from artist Susan Morris's motion capture drawing practice. In her work, Morris has

explored how movement can be algorithmically translated into complex, often abstract visualisations, such as fog-like drawings that index the body's presence over time, rather than represent it figuratively. Her work demonstrates that motion capture is not merely a recording device, but a temporal and gestural medium, capable of revealing what she calls the "traces of involuntary movement" including actions we may not even be aware we are performing [Morris, 2012]. This notion resonated strongly with our approach, as students engaged in improvisational movement that exceeded deliberate control, producing forms shaped as much by spontaneity, proximity, and relational gestures as by any planned outcome.

Similarly, in our workshop, the unpredictable alignment of analogue materials such as charcoal with digital motion capture technology created novel outcomes, expanding the scope of what could be considered a 'drawing.' Students' movements were not an end in themselves but a generative source for visual outcomes, including layered motion drawings, time-based video works, and collaborative large-scale paper pieces. By connecting embodied action to drawn traces, we reinforced the idea that physical movement itself can be a form of mark making and creative expression. The compiled video and motion-generated images served as tangible artefacts linking the ephemeral workshop activities to concrete artistic results, paralleling Morris's translation of performance into visual form. These outcomes provided a platform for critical discussion on how gesture, temporality, and mediation intersect in contemporary drawing. They also gave students a sense of authorship, as each participant could identify their contribution within the collective piece. This underlined how their embodied engagement and playful improvisation culminated in a meaningful artistic product that blurred the line between performance, data, and drawing.

6 MIXED METHODS EVALUATION

A mixed-methods approach was used to evaluate the workshop. A post-session survey, distributed via Google Forms, included sixteen Likert-scale questions and one open-ended question, exploring engagement, collaboration, and risk-taking. The quantitative questions were analysed using descriptive statistics. The responses to the open-ended questions in the survey were thematically coded using deductive analysis [Saldaña 2013]. Qualitative observations, still images, and motion-capture artefacts supplemented survey data to provide a multimodal understanding of student experience. Using content analysis [Krippendorff, 2019], we looked at photographic and video documentation to explore student gestures, group dynamics, and the drawing and motion capture outcomes. This multimodal perspective helped contextualise students' written reflections. We are going to discuss the results in relation to our research objectives.

The survey investigated the experience of the workshop, the communication effectiveness of the workshop and the effect of technology on innovation. In general, students who attended the workshop found it to be engaging and an innovative learning experience. Most of them identified that it was "enjoyable" or "very enjoyable." They enjoyed the playful interaction the most as they regard these playful and improvisational activities as helpful in their understanding of drawing as a more process-based and time-based practice that is collaborative. Among all the challenges, they found



Figure 2: Image showing an example of a hybrid drawing, motion capture and video artefact of the workshop used in the content analysis

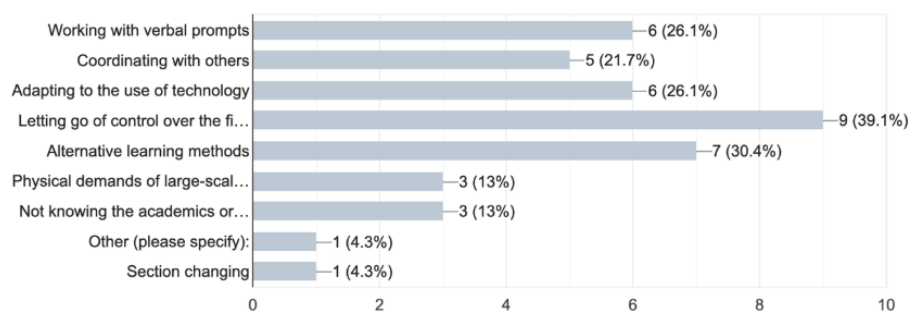


Figure 3: The most challenging aspect of the workshop.

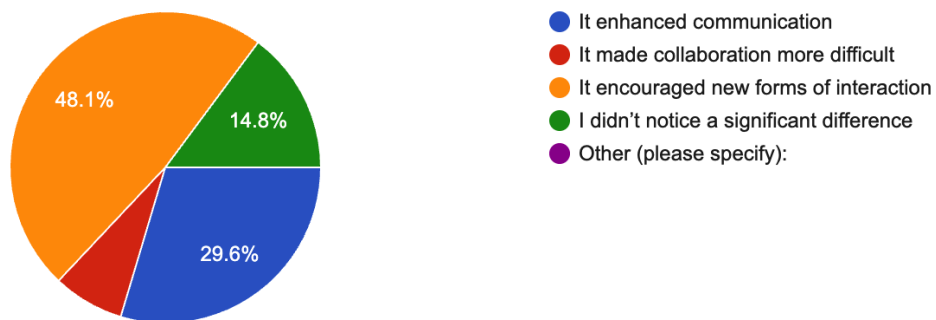


Figure 4: Participants' feelings about verbal prompts

the most challenging part is to let go of controlling the outcome and accepting the unpredictable results (Figure 3).

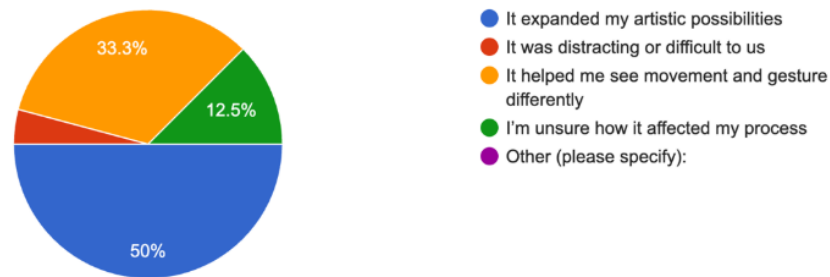
Most of the participants feel the verbal prompts and level of freedom in the workshop enhance communication and encourage new forms of interaction (Figure 4). Half of the participants agree

that the workshop creates a sense of connection or shared purpose with other participants. They are more motivated by the feeling of pursuing a collaborative goal.

The student responses were thematically analysed using Braun and Clarke's [2006] six-phase thematic analysis method. Data was

Table 1: Selective data from selected themes to show representative responses.

Theme	Sub-theme	Coding unit
Play	Skills or knowledge of drawing or motion capture Unknown or new method	"The event was great. It was a new experience for me. Overall, it made me realize some things I had never thought of before, and showed me that there are many ways to create art, that it can be very free and fun."
Play	Embodied awareness	"That's a very interesting experience of lying of the ground and getting captured. The workshop also makes me have a deeper understanding of the motion capture."
Play	Embodied awareness	"I'm happy that we get to interact instead of sitting and just listening"
Engagement	Unknown or new method	"Letting go of control over the final artwork pushed me creatively. Unpredictable, but a great lesson in flexibility!"
Engagement	Embodied awareness Unknown or new method	"I shifted from drawing only what I see to using emotional, but this activities forced me to feel movement first, then translate it into lines. It is a totally new approach for me."
Engagement	Experimentation Unknown or new method	"Drawing something meaningless is harder than I imagined. We are too used to drawing known things rather than the unknown..."
Engagement	Skills or knowledge of drawing or motion capture	"Never thought about how motion capture could work with drawing"
Collaboration	Skills or knowledge of drawing or motion capture	"I enjoyed the collaborative drawing activities, bouncing ideas off peers and random words made everything more creative."
Collaboration	Skills or knowledge of drawing or motion capture	"This event includes interaction with other so we are more focused on the topic."
Collaboration	Skills or knowledge of drawing or motion capture	"Able to enjoy interaction with others during gaining knowledge to new technologies"

**Figure 5: The influence of technology integration on the creative process**

first reduced into questions, then were deductively coded for recurring themes related to our research questions on engagement, collaboration, and play. Codes were then grouped into sub themes, such as "skills or knowledge of drawing or motion capture", "embodied awareness", "experimentation" and "freedom." These themes were refined through cross-checking by two researchers to ensure interpretive consistency. Representative quotes from three themes and sub-themes were selected to illustrate a sample of the results. Table 1 below shows these select number of quotes, highlighting how students linked "play," collaboration," "embodied awareness," "drawing" and "motion capture" to the workshop.

The results show that the innovative use of motion capture technology in artistic collaboration was intriguing, as most of them used to think that it was used only in industrial workflows rather than creative processes. They stated that the workshop reshaped their approaches to creating in a more expressive, gesture-based way and a boundary-less approach, which encourages creative risk-taking, spontaneous decision-making and openness to unexpected outcomes. (Figure 5)

7 Feedback Analysis and Discussion

The feedback confirms that the workshop's focus on embodied learning, multimodal interaction, and shared authorship effectively

fostered a positive emotional atmosphere and a reflective learning space. This outcome supports theories of embodied cognition which argue that knowledge emerges through sensorimotor engagement and situated practice [Barsalou, 2008; Wilson, 2002]. The emphasis on novelty in responses underscores the importance of framing motion capture as part of an experimental process rather than a final product, valuing process and iteration above outputs [Sullivan, 2010]. This perspective encouraged participants to experiment more, take risks [Veon, 2023], and physically engage with drawing, aligning with the workshop's aim to stimulate embodied cognition and multimodal enrichment. This resonates with Mathias and von Kriegstein's [2023] concept of multimodal enrichment, where learning is deepened through the activation of multiple sensory and motor channels. Participants' attitudes toward accepting the new learning experience demonstrated openness and an embrace of the innovative approach with enthusiasm. They were willing to step outside their comfort zones and explore an unfamiliar style of learning in an environment that required them to think differently. Importantly, the drawing activities generated a sense of joy and inclusion that helped students feel more relaxed. This collective enjoyment reduced anxiety, lowered the fear of failure, and encouraged risk-taking, a dynamic consistent with Universal Design for Learning's emphasis on engagement through play and relevance [CAST, 2018; Forbes, 2021]. By fostering belonging and safety in the group, the workshop enabled students to move beyond risk-aversion and into more adventurous forms of creative exploration. The participants never expected to be ready to engage with advancing techniques and technologies, highlighting their willingness to learn and expand their conceptions of drawing. This newfound confidence allowed them to embrace challenges and experiment with techniques they previously thought were beyond their abilities. By the end of the workshop, participants not only improved their skills but also gained a deeper understanding of how creativity and collaboration enhance learning. Their openness not only enriched their personal learning but also fostered a collaborative and energetic group environment, leading to a better understanding of the workshop's goals.

Furthermore, the workshop created the conditions for what Csikszentmihalyi [1990] describes as "flow," where participants became fully absorbed in creative risk-taking without self-consciousness. In this sense, the workshop provided not just enjoyment but also fostered playful, embodied engagement. This unexpected reorientation mirrors Ratto's [2011] notion of "critical making," where hands-on experimentation provokes new perspectives and reflective understanding. Their openness not only enriched personal learning but also fostered a collaborative and energetic group environment, leading to a better grasp of the workshop's goals. This approach encourages learners to interact directly with materials and tools, sparking new insights and promoting a deeper understanding of the subject. By combining critical thinking with creative practice, participants develop the ability to question assumptions and explore innovative solutions.

The challenges identified in the questionnaire responses, such as collaborating with others, switching quickly between activities, and letting go of personal control, closely align with the pedagogical goal of encouraging students to move beyond habitual, individual

approaches. They became engrossed as they concentrated on completing the tasks, unaware of the outcome. The fact that participants frequently viewed these challenges as constructive indicates that the workshop effectively promoted adaptability. The responses further reveal an increasing awareness of the posthuman and relational dimensions of authorship in technology-mediated collaboration: students observed that the outcomes were shaped not just by their individual input but also by interactions among peers, facilitators, drawing tools, motion capture devices, and other computational and analogue components. This relational perspective resonates with Bennett's [2010] notion of "vibrant matter" and Manning's [2009] framing of creativity as emerging from assemblages of human and nonhuman actors. It highlights how collaboration and adaptability can produce innovative results driven by the combination of different influences.

In summary, the questionnaire results indicate that the workshop achieved its objectives: redefining drawing as an embodied and continuous activity, fostering critical engagement with digital materiality, and providing conditions where students could act as co-creators in a shared creative process. The data also points to potential improvements, such as extending the activity duration slightly and incorporating structured moments for intergroup exchange, that could enhance dialogue among participants without compromising the openness and fluidity they appreciated.

8 CONCLUSION

This relational view of authorship is deeply aligned with the DRG's pedagogical ethos, which resists traditional notions of drawing as the outcome of a singular, masterful hand. Instead, the group embraces drawing as a dynamic, open-ended, and performative process, one that is shaped by materials, tools, environments, and collaborative energies as much as by individual intent. This paper has shown how a collaborative workshop integrating experimental motion capture and drawing is reframed as an embodied, multimodal, and relational practice. By treating motion capture as a generative material rather than a precision recording tool, the workshop expanded drawing into a full-body, spatial, and time-based process, activating embodied learning and multimodal thinking. Unpredictable moments of disruption proved productive forces for creativity. This echoes Bishop's [2012] view that antagonism in participatory art generates new meanings, moments of friction, or uncertainty. Rather than errors or tension, they were reframed as catalysts for collaboration and dialogue, and opportunities for innovation and reflection. The workshop also reconfigured authorship in a posthuman context. Creativity emerged as distributed across students, facilitators, materials, and technologies, dismantling the notion of the singular artist and positioning drawing as an event shaped by human and nonhuman actors alike. Findings further highlight the value of experimental learning literacy fostered through play, technology, and sensory engagement. The multimodal setting encouraged agency, inclusivity, and digital fluency, modelling an educational ecology where risk-taking and collaboration are central. This study contributes to emerging conversations around multimodal, inclusive pedagogy by explicitly contrasting process-based and outcome-based learning models. The findings demonstrate that embodied, collaborative frameworks not only

improve engagement but also cultivate critical reflection on technology and authorship. By identifying gaps in conventional art and animation education, particularly its emphasis on polished output, the project evidences how playful disruption can be reimagined as a rigorous mode of inquiry. These outcomes underscore the significance of embodied, interdisciplinary practice for art and design education in an era of digital and post-disciplinary creativity. The workshop points toward a pedagogy that is inclusive, experimental, and future-facing, one that embraces disruption as a generative condition for learning. In the end, it did more than stage a drawing session or capture motion, it revealed learning itself as a living, collaborative art, ever unfinished and ever evolving.

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